

MOHAMMAD MERAJ

Senior Generative AI Engineer | Lead Data Scientist

Agentic AI | RAG & LLM Applications | Machine Learning | LLMOps & MLOps | AWS & Azure

Bangalore, India

meet.meraj2000@gmail.com | +91-7019709092 | [linkedin.com/in/mdmeraj](https://www.linkedin.com/in/mdmeraj)



PROFESSIONAL SUMMARY

Senior AI/ML professional with 12 years of enterprise technology experience, evolving from analytics and Data Science into hands-on development of Generative AI, Agentic AI, RAG and LLM applications. Designs, builds and deploys production-grade multi-agent systems and machine-learning solutions using Python, LangGraph, LangChain, FastAPI, AWS/Azure and vector databases, with LLMOps and MLOps practices for observability and CI/CD. Strong Data Science foundation in predictive modelling, NLP, clustering and anomaly detection. Combines hands-on engineering with technical ownership, cross-functional collaboration and client engagement across BFSI, insurance and cybersecurity. Pursuing a PhD in Data Science and AI.

CORE SKILLS

Generative & Agentic AI: Agentic AI, Multi-Agent Systems, LangGraph, LangChain, CrewAI, n8n, MCP, RAG, AWS Bedrock, Hugging Face, LLaMA, Gemma, Groq, Prompt Engineering, Human-in-the-Loop

Machine Learning, Deep Learning & Data Science: Regression, Classification, Predictive Modelling, NLP, XGBoost, Random Forest, Gradient Boosting, Decision Trees, K-Means Clustering, Isolation Forest, Anomaly Detection, Customer Segmentation, Computer Vision, Object Detection, Domain Adaptation, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Statistical Analysis

Programming Languages: Python, PySpark, SQL

Databases & Vector DBs: PostgreSQL, NoSQL, Pinecone, ChromaDB, FAISS, pgvector, Qdrant

Cloud: AWS (Bedrock, SageMaker, Lambda, S3, EC2, ECS, ECR, Glue, IAM, API Gateway), Azure (Databricks, ML Studio)

LLMOps & MLOps: Langfuse, Grafana, GitHub Actions, CI/CD, Docker, Git, Linux

Backend Development: FastAPI, Flask, Django, Streamlit, REST APIs

Leadership & Delivery: Technical Ownership, Client & Stakeholder Engagement, Agile/SDLC, Team Leadership, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

AI Engineer Advisor | NTT DATA

Apr 2025 – Present

Generative AI & Agentic AI Engineering

Panasonic Well – PRD-to-Jira Multi-Agent Platform

- Designed and developed a production-oriented multi-agent platform using Python, LangGraph, LangChain and FastAPI that converts product requirement documents (PRDs) into structured, auditable Jira work items.
- Built two LLM agents: a Completeness Auditor Agent that scores PRDs on coverage, clarity, traceability and testability, and an Extraction Agent that converts validated PRDs into hierarchical Epic/Story JSON.
- Implemented a human-in-the-loop workflow with a React-based interface, enabling users to review, edit and approve generated Epics and Stories before Jira submission, backed by FastAPI services and an auditable workflow.
- Integrated Google Drive PRD ingestion, Jira REST APIs (Epic/Story creation, linking and status transitions) and Slack notifications into the pipeline.
- Implemented LLMOps practices with Docker, GitHub Actions CI/CD and Langfuse/Grafana observability, with Pinecone for vector storage, supporting reliable deployment and monitoring of the multi-agent workflow. Impact: reduced manual PRD processing time and improved documentation quality.

NTT DATA Internal – Multi-Agent Stock Recommendation

- Built an agentic RAG application in Python and LangChain that answers plain-English questions over historical and real-time stock data stored in PostgreSQL.
- Developed a Text-to-SQL agent using LangChain create_sql_agent to generate and execute SQL that extracts and aggregates tabular financial data.
- Implemented a recommendation agent combining structured SQL results with ChromaDB semantic retrieval (sentence-transformer embeddings of stock metadata) and GPT-4 reasoning to deliver explainable, context-aware recommendations.

GT UK Audit Team – RAG Document Q&A

- Developed a financial-document Q&A application for the GT UK Audit Team using Retrieval-Augmented Generation to answer domain-specific audit and finance questions.
- Engineered the RAG workflow in Python with LangChain, embedding financial documents, retrieving relevant context for each question and generating document-grounded answers.
- Delivered the application end to end on AWS: Bedrock for model inference, S3 for document storage, Lambda for serverless processing and IAM for secure access, with a Streamlit user interface.

UK Insurance Client – GenAI Call Summarization

- Built a Python workflow that transcribes customer service calls with Whisper AI and generates concise, actionable summaries using Gemma via the Groq API.
- Refined and tuned the model setup to reduce hallucinations and improve sentiment-analysis precision in the generated output.
- Delivered the flow from customer call to transcript, summary and sentiment output through a Streamlit user interface.

Senior Consultant, Lead Data Scientist | Capgemini

Mar 2021 – Apr 2024

Capgemini Internal – Email Categorization with Generative AI

- Developed a Generative AI prototype to automatically categorize emails into predefined categories such as Politics, Sports, Technology and Entertainment using Python, LangChain and Hugging Face Transformers.
- Collected and pre-processed a large dataset of news articles from multiple sources to support the categorization workflow, with Flask used for the application layer.

Standard Chartered Bank, Singapore – Customer Segmentation

- Analyzed customer payment and behavioral data to identify distinct customer segments and support targeted cross-selling of relevant banking products and services.
- Developed customer segmentation and predictive models using clustering, XGBoost and Random Forest in Python and PySpark to identify behavioral patterns and customer groups with different product needs.
- Deployed the solution on AWS using EC2, ECS and S3, enabling the segmentation models to support production-driven customer targeting and campaign execution.
- Enabled data-driven cross-selling and targeted customer engagement through model-generated segments, contributing to a 4% increase in sales for the bank.

Capgemini France – SOC Anomaly Detection

- Developed an AI-driven anomaly-detection solution for a cybersecurity SOC team using Isolation Forest and K-Means on Azure Databricks and Azure ML Studio, automating anomaly identification and reducing manual intervention.
- Deployed and supported the solution in Azure production, with a user-feedback loop that dynamically improved the model, and analyzed production issues to deliver fixes within SLA.

Freelance Data Scientist

Sep 2019 – Mar 2021

- Developed propensity-to-pay models using Decision Trees and Gradient Boosting to identify high-potential customers and support targeted customer outreach.
- Applied K-Means clustering to segment customers based on lifestyle and life-stage characteristics, enabling more targeted customer engagement strategies.
- Analyzed control vs. experiment groups and historical customer behavior to optimize campaign timing, customer journeys and lead-generation strategies.

Advanced Analyst | EY

Sep 2013 – Nov 2017

Deutsche Bank – Financial Analysis & Reporting

- Performed financial data and data-quality analysis for Deutsche Bank's global operations, ensuring compliance with accounting standards and regulatory frameworks, and prepared reports and dashboards for senior management.
- Led a team by assigning tasks and tracking progress to deliver timely, high-quality work, with daily and monthly status reporting to stakeholders.

EDUCATION

PhD, Data Science and AI – CMR University (Pursuing) 2025 – 2029*Research: domain adaptation for robust object detection in adverse driving conditions.***MSc, Data Science** – Chandigarh University 2021 – 2023**BSc, IGNOU** – Govt. of India 2003 – 2006**CERTIFICATIONS & PROFESSIONAL DEVELOPMENT**

Azure AI Fundamentals (AI-900), Microsoft | MLOps Workshop, Capgemini